

Scenario 1: Customer Satisfaction Analysis

Question:

A company wants to analyze customer satisfaction survey data. How can Data Science concepts help to clean and analyze this data?

- Collect customer satisfaction survey data create a dataframe
- Load the data using pandas
- Understand the data structure, columns, and rating scales
- Identify missing, duplicate, or incorrect values.
- Fill missing values (mean/median for numerical, mode for categorical).
- Remove duplicate survey responses.
- Fix inconsistent data (standardize text, rating formats).
- Detect and handle outliers in satisfaction scores.
- Convert categorical data into numerical form (encoding).
- Normalize or scale numerical features if required.
- Perform exploratory data analysis (average scores, trends, patterns).
- Analyze satisfaction across products, regions, or customer types.
- Process text feedback using sentiment analysis (if available).
- Identify key factors affecting customer satisfaction.
- Apply clustering to group customers by satisfaction behavior.
- Build predictive models to predict satisfaction levels (optional).
- Evaluate model performance and validate results.
- Interpret findings to generate business insights.
- Visualize results using charts and dashboards.
- Present insights to stakeholders for decision-making.

Scenario 2: Product Sales Performance

Question:

A retail store has recorded monthly sales data for its products. How can Data Science concepts be applied to understand sales performance?

- Create a dataframe with sales , date, and product details
- Calculate the frequency of each products purchase for that month
- Calculate the total cost of the product for each month determine the which month the sales is higher for which product
- Use the bar chart to visualize the sales for that particular product.
- Use line chart to analyze the sales trends for each of the products.

Scenario 3: Medical Test Effectiveness

Question:

A hospital conducted a study to check the effectiveness of a new medical test. How can Data Science concepts be used to validate its results?

We can split the dataset into groups then perform ttest for the each of the groups to detect the p value. From the p value we can determine whether the medicine is effective or not effective.

Scenario 4: Employee Performance Evaluation

Question:

A company wants to evaluate employee performance using various metrics. How can Data Science concepts be applied?

- Collect employee data (attendance, sales, targets achieved, feedback, KPIs)
- Clean and organize the data for accuracy
- Use descriptive analytics to summarize performance (averages, trends)
- Apply performance metrics and scoring models
- Analyze the rating of each employee to calculate the performance
- Calculate the employee performance across each project.
- Determine which project has highest employee performance
- Use predictive analytics to forecast future performance
- Identify top performers and employees needing support
- Visualize results using charts and dashboards
- Support data-driven decisions for promotions, training, and rewards

Scenario 5: Flight Delay Investigation

Question:

An airline tracks flight delays (in minutes) for 30 flights. How can Data Science concepts help in understanding the delays?

- Create a pandas dataframe
- Calculate Q1 `np.percentile(data, 25)` 25th percentile
- Calculate Q3 `np.percentile(data, 75)` 75th percentile
- Calculate interquartile range $IQR = Q3 - Q1$
- Calculate the lower and upper bound ranges
- Lower bound = $Q1 - 1.5 IQR$ Upper bound = $Q3 + 1.5 IQR$
- If value is lesser than lower bound then it is an lesser outlier